Darshan Valentino Prasad

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PROFESSIONAL SUMMARY

Computer Science (Data Science) undergraduate with hands-on experience developing end-to-end ML systems, multi-modal retrieval pipelines, and analytics solutions. Built production-style projects involving FastAPI deployment, RAG architectures, computer vision, and predictive modeling using datasets of up to 120K samples. Interested in applied ML, data science, AI infrastructure, and MLOps.

TECHNICAL SKILLS

Programming: Python, SQL, Bash
ML Algorithms: Scikit-learn, XGBoost, LightGBM
Deep Learning: PyTorch, Transformers, TorchVision, Hugging Face
LLMs & Generative AI: RAG Systems, LangGraph, Sentence Transformers, BLIP-2, Prompt Engineering
Data Engineering: Pandas, NumPy, OCR Pipelines, Feature Engineering, ETL
Databases & Vector Search: PostgreSQL, MongoDB, FAISS-based vector search
Backend & Deployment: FastAPI, REST APIs, Docker
Analytics & Visualization: Power BI, DAX, Plotly, Excel
Tools: Git, Linux

PROJECTS

AWARE - Adaptive Wearable Analytics for glucose Risk Estimation Sept 2026

- Developed the predictive framework to analyze multi-channel wearable physiological data, enabling continuous forecasting of critical health metrics.
- Processed and evaluated extensive real-world datasets (e.g., PhysioNet), achieving high prediction accuracy with a Mean Absolute Error (MAE) of 15.27 mg/dL and a Pearson correlation of 0.81.
- Validated clinical safety and reliability using the Clarke Error Grid, successfully placing 100% of predictions within the clinically acceptable Zones A (62.5%) and B (37.5%).
- Profiled and optimized the end-to-end data processing pipeline for edge deployment, achieving an inference latency of 0.023 ms and a throughput of over 42,700 samples per second.
- **Tech Stack:** Python, PyTorch, XGBoost, Scikit-Learn, Predictive Modeling, Data Visualization

Geo-Caster: Autonomous Disaster Prediction Platform April 2026
Geo-Caster GitHub Repository

- Developed an end-to-end modular ML pipeline supporting reproducible training, automated feature engineering, model evaluation, and production inference.
- Built a preprocessing pipeline capable of handling 120k+ observations with satellite-derived environmental features through modular ETL stages.
- Engineered 40+ climate and satellite features.
- Achieved 82% test coverage across preprocessing, feature engineering, and inference modules using pytest.
- Improved ROC-AUC from 0.91 to 0.9725 while reducing false positives by 34%.
- Deployed a production inference service handling concurrent prediction requests through FastAPI.
- Built an observability dashboard tracking inference latency, prediction distribution, model drift, and retraining history.
- **Tech Stack:** Python, Pandas, NumPy, scikit-learn, FastAPI, Plotly, Pytest

Intelligent RAG & VLM Retrieval Platform Oct 2025
LYRA GitHub Repository

- Independently built an end-to-end RAG pipeline using OCR, BLIP-2, Sentence Transformers, and FAISS, supporting semantic retrieval across scanned PDFs.
- Indexed 18,000 document chunks using FAISS, achieving an average retrieval latency below 180 ms.
- Reduced hallucinations by evaluating chunk size and embedding strategies across multiple retrieval configurations.
- Benchmarked MiniLM, BGE-large, and E5 embeddings across Precision@1, Recall@5, and MRR, selecting BGE-large after a 14% retrieval improvement.
- Containerized a FastAPI application using Docker for reproducible deployment.
- Generated traceable, coherent responses using a local Mistral 2B LLM.
- **Tech Stack:** Python, FastAPI, PyTorch, FAISS, Sentence Transformers, BLIP-2, EasyOCR

EDUCATION

Bachelor of Engineering in Computer Science (Data Science) 2023 – Present
Jawaharlal Nehru New College of Engineering, Shivamogga CGPA: 8.93

Leadership & Activities

- Student Coordinator, Datorum – Official Data Science Forum, JNNCE
- Core Member, Tech Team – Placement and Training Office, JNNCE
- Former Technical Team Lead, ByteBrigade, JNNCE
- Recipient of the Best Project Award, Second Semester, JNNCE

Pre-University Course (PUC) 2021 – 2023
PACE PU College, Shivamogga 83.83%

10th Grade 2021
Podar International School, Shivamogga 88.83%

CERTIFICATIONS

- MongoDB for Students, MongoDB University — 13 July 2026
- Japanese Language Proficiency Certificate (N5 Certification), Zenken — Nov 10, 2025
- Artificial Intelligence Fundamentals, IBM SkillsBuild — Oct 29, 2025
- Machine Learning, MasterCampus Academy — 27 Sep 2025
- CS50x, Harvard Online — 2025
- Microsoft Power BI, MasterCampus Academy — Feb 22, 2025

Technical Achievements

- Patent applied for AWARE's predictive framework
- Python Instructor, The Coding Bootcamp, Tech Team (T&P Department), JNNCE — July 2026
- Lead Project Maintainer & Architect, JNNCE Legacy System — July 2026
  Project Repository: Legacy System GitHub Repository
- National Hackathon Finalist, LuminusX, RNSIT Bangalore — April 2026
- Technical Head, HackFest-1.0 (State-Level Hackathon), ByteBrigade Club, JNNCE — Mar 2026
- Event Coordination Volunteer, TechZone 2k25 — Nov 2025
- Event Coordinator, TechZone 2k25 — May 2025

LANGUAGES

English, Hindi, Kannada, Konkani, Japanese (Beginner Proficiency)